

Name: Sagi Sahasra Srihasa

Roll no: 2520030008

Step 1:

```
1  /* Name: Sagi Sahasra Srihasa
2     Roll no: 2520030008
3  */
4
5  -- Step 1: Create Database and Table
6  CREATE DATABASE orders_management_db;
7  USE orders_management_db;
8  CREATE TABLE orders (
9      ord_no INT PRIMARY KEY,
10     purch_amt DECIMAL(10,2),
11     ord_date DATE,
12     customer_id INT,
13     salesman_id INT
14 );
```

30	18:13:38	CREATE DATABASE orders_management_db	1 row(s) affected
31	18:13:41	USE orders_management_db	0 row(s) affected
32	18:13:43	CREATE TABLE orders (ord_no INT PRIMARY KEY, purch_amt DECIMAL(10,2), ord_date DATE, ...	0 row(s) affected

Step 2:

```
-- Step 2: Insert Data
INSERT INTO orders VALUES
(70001,150.50,'2012-10-05',3005,5002),
(70009,270.65,'2012-09-10',3001,5005),
(70002,65.26,'2012-10-05',3002,5001),
(70004,110.50,'2012-08-17',3009,5003),
(70007,948.50,'2012-09-10',3005,5002),
(70005,2400.60,'2012-07-27',3007,5001),
(70008,5760.00,'2012-09-10',3002,5001),
(70010,1983.43,'2012-10-10',3004,5006),
(70003,2480.40,'2012-10-10',3009,5003),
(70012,250.45,'2012-06-27',3008,5002),
(70011,75.29,'2012-08-17',3003,5007),
(70013,3045.60,'2012-04-25',3002,5001);
```

33	18:14:39	INSERT INTO orders VALUES (70001,150.50,'2012-10-05',3005,5002), (70009,270.65,'2012-09-10',3001,500...	12 row(s) affected Records: 12
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Step 3:

```
29 -- Step 3: WHERE Clause
30 • SELECT *
31 FROM orders
32 WHERE purch_amt > 2000;
33 -- Step 4: WHERE Clause with Date
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content: |

ord_no	purch_amt	ord_date	customer_id	salesman_id
70003	2480.40	2012-10-10	3009	5003
70005	2400.60	2012-07-27	3007	5001
70008	5760.00	2012-09-10	3002	5001
70013	3045.60	2012-04-25	3002	5001
NULL	NULL	NULL	NULL	NULL

35 18:50:30 SELECT * FROM orders WHERE purch_amt > 2000 LIMIT 0, 1000

Step 4:

```
33 -- Step 4: WHERE Clause with Date
34 • SELECT *
35 FROM orders
36 WHERE ord_date = '2012-09-10';
37
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content: |

ord_no	purch_amt	ord_date	customer_id	salesman_id
70007	948.50	2012-09-10	3005	5002
70008	5760.00	2012-09-10	3002	5001
70009	270.65	2012-09-10	3001	5005
NULL	NULL	NULL	NULL	NULL

Step 5:

```
39 -- Step 5: WHERE Clause with Salesman
40 • SELECT *
41 FROM orders
42 WHERE salesman_id = 5001;
43 -- Step 6: ORDER BY Purchase Amount
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content: |

ord_no	purch_amt	ord_date	customer_id	salesman_id
70002	65.26	2012-10-05	3002	5001
70005	2400.60	2012-07-27	3007	5001
70008	5760.00	2012-09-10	3002	5001
70013	3045.60	2012-04-25	3002	5001
NULL	NULL	NULL	NULL	NULL

Step 6:

```

43  -- Step 6: ORDER BY Purchase Amount
44  • SELECT *
45  FROM orders
46  ORDER BY purch_amt DESC;
47  -- Step 7: ORDER BY Order Date

```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content: [IA](#)

ord_no	purch_amt	ord_date	customer_id	salesman_id
70008	5760.00	2012-09-10	3002	5001
70013	3045.60	2012-04-25	3002	5001
70003	2480.40	2012-10-10	3009	5003
70005	2400.60	2012-07-27	3007	5001
70010	1983.43	2012-10-10	3004	5006
70007	948.50	2012-09-10	3005	5002
70009	270.65	2012-09-10	3001	5005
70012	250.45	2012-06-27	3008	5002
70001	150.50	2012-10-05	3005	5002
70004	110.50	2012-08-17	3009	5003
70011	75.29	2012-08-17	3003	5007
70002	65.26	2012-10-05	3002	5001
NULL	NULL	NULL	NULL	NULL

Step 7:

```

47  -- Step 7: ORDER BY Order Date
48  • SELECT *
49  FROM orders
50  ORDER BY ord_date ASC;
51  -- Step 8: SUM()

```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content: [IA](#)

ord_no	purch_amt	ord_date	customer_id	salesman_id
70013	3045.60	2012-04-25	3002	5001
70012	250.45	2012-06-27	3008	5002
70005	2400.60	2012-07-27	3007	5001
70004	110.50	2012-08-17	3009	5003
70011	75.29	2012-08-17	3003	5007
70007	948.50	2012-09-10	3005	5002
70008	5760.00	2012-09-10	3002	5001
70009	270.65	2012-09-10	3001	5005
70001	150.50	2012-10-05	3005	5002
70002	65.26	2012-10-05	3002	5001
70003	2480.40	2012-10-10	3009	5003
70010	1983.43	2012-10-10	3004	5006
NULL	NULL	NULL	NULL	NULL

Step 8:

```
51 -- Step 8: SUM()
52 • SELECT SUM(purch_amt) AS total_revenue
53 FROM orders;
54 -- Step 9: AVG()
```

Result Grid	Filter Rows:	Export:	Wrap Cell Content:
total_revenue			
▶ 17541.18			

Step 9:

```
54 -- Step 9: AVG()
55 • SELECT AVG(purch_amt) AS average_order
56 FROM orders;
57 -- Step 10: MAX()
```

Result Grid	Filter Rows:	Export:	Wrap Cell Content:
average_order			
▶ 1461.765000			

Step 10:

```
57 -- Step 10: MAX()
58 • SELECT MAX(purch_amt) AS highest_order
59 FROM orders;
60 -- Step 11: MIN()
```

Result Grid	Filter Rows:	Export:	Wrap Cell Content:
highest_order			
▶ 5760.00			

Step 11:

```
60 -- Step 11: MIN()
61 • SELECT MIN(purch_amt) AS lowest_order
62 FROM orders;
63 -- Step 12: COUNT()
```

Result Grid	Filter Rows:	Export:	Wrap Cell Content:
lowest_order			
▶ 65.26			

Step 12:

```
63 -- Step 12: COUNT()
64 • SELECT COUNT(*) AS total_orders
65 FROM orders;
66 -- Step 13: GROUP BY Salesman
```

Result Grid			Filter Rows:		Export:		Wrap Cell Content:	
	total_orders							
▶	12							

Step 13:

```
66 -- Step 13: GROUP BY Salesman
67 • SELECT salesman_id,
68         SUM(purch_amt) AS total_sales
69 FROM orders
70 GROUP BY salesman_id;
```

Result Grid			Filter Rows:		Export:		Wrap Cell Content:	
	salesman_id	total_sales						
▶	5002	1349.45						
	5001	11271.46						
	5003	2590.90						
	5005	270.65						
	5006	1983.43						
	5007	75.29						

Step 14:

```
73 -- Step 14: GROUP BY Customer
74 • SELECT customer_id,
75         SUM(purch_amt) AS total_purchase
76 FROM orders
77 GROUP BY customer_id;
```

Result Grid			Filter Rows:		Export:		Wrap Cell Content:	
	customer_id	total_purchase						
▶	3005	1099.00						
	3002	8870.86						
	3009	2590.90						
	3007	2400.60						
	3001	270.65						
	3004	1983.43						
	3003	75.29						
	3008	250.45						

Step 15

```
80  -- Step 15: GROUP BY + MAX()
81  •  SELECT customer_id,
82         MAX(purch_amt) AS highest_purchase
83  FROM orders
84  GROUP BY customer_id;
```

Result Grid	Filter Rows:	Export:	Wrap Cell Content:
customer_id	highest_purchase		
3005	948.50		
3002	5760.00		
3009	2480.40		
3007	2400.60		
3001	270.65		
3004	1983.43		
3003	75.29		
3008	250.45		

Step 16:

```
87  -- Step 16: GROUP BY + HAVING
88  •  SELECT salesman_id,
89         SUM(purch_amt) AS total_sales
90  FROM orders
91  GROUP BY salesman_id
92  HAVING SUM(purch_amt) > 3000;
```

Result Grid	Filter Rows:	Export:	Wrap Cell Content:
salesman_id	total_sales		
5001	11271.46		

Step 17:

```
95  -- Step 17: GROUP BY + HAVING
96  •  SELECT customer_id,
97         SUM(purch_amt) AS total_purchase
98  FROM orders
99  GROUP BY customer_id
100  HAVING SUM(purch_amt) > 2500;
101
```

Result Grid	Filter Rows:	Export:	Wrap Cell Content:
customer_id	total_purchase		
3002	8870.86		
3009	2590.90		

Step 18:

```
104 -- Step 18: GROUP BY + HAVING
105 • SELECT customer_id,
106         COUNT(*) AS total_orders
107 FROM orders
108 GROUP BY customer_id
109 HAVING COUNT(*) > 1;
110 -- Step 19: GROUP BY + HAVING + ORDER BY
```

Result Grid |  Filter Rows: | Export:  | Wrap Cell Content: 

	customer_id	total_orders
▶	3005	2
	3002	3
	3009	2

Step 19:

```
110 -- Step 19: GROUP BY + HAVING + ORDER BY
111 • SELECT customer_id,
112         SUM(purch_amt) AS total_purchase
113 FROM orders
114 GROUP BY customer_id
115 HAVING SUM(purch_amt) > 1000
116 ORDER BY total_purchase DESC;
117 -- Step 20: BETWEEN + HAVING
```

Result Grid |  Filter Rows: | Export:  | Wrap Cell Content: 

	customer_id	total_purchase
▶	3002	8870.86
	3009	2590.90
	3007	2400.60
	3004	1983.43
	3005	1099.00

Step 20:

```
117 -- Step 20: BETWEEN + HAVING
118 • SELECT customer_id,
119         MAX(purch_amt) AS max_purchase
120 FROM orders
121 GROUP BY customer_id
122 HAVING MAX(purch_amt) BETWEEN 2000 AND 6000;
123 -- Step 21: COUNT + HAVING
```

Result Grid |  Filter Rows: | Export:  | Wrap Cell Contents: 

	customer_id	max_purchase
▶	3002	5760.00
	3009	2480.40
	3007	2400.60

Step 21:

```
123 -- Step 21: COUNT + HAVING
124 • SELECT salesman_id,
125         COUNT(*) AS total_orders
126 FROM orders
127 GROUP BY salesman_id
128 HAVING COUNT(*) >= 2;
129 -- Step 22: MAX + GROUP BY + HAVING
```

Result Grid	Filter Rows:	Export:	Wrap Cell Content:
salesman_id	total_orders		
5002	3		
5001	4		
5003	2		

Step 22:

```
129 -- Step 22: MAX + GROUP BY + HAVING
130 • SELECT ord_date,
131         MAX(purch_amt) AS highest_purchase
132 FROM orders
133 GROUP BY ord_date
134 HAVING MAX(purch_amt) > 2000;
```

Result Grid	Filter Rows:	Export:	Wrap Cell Content:
ord_date	highest_purchase		
2012-10-10	2480.40		
2012-07-27	2400.60		
2012-09-10	5760.00		
2012-04-25	3045.60		